Q. No.	Questions
UNIT 1	**SECTION 1**
1	An organization is deciding how to connect its two branch offices located 1500 km apart. Option 1 uses a dedicated leased line directly between the two offices. Option 2 connects both offices to a shared backbone network made up of multiple point-to-point WANs joined by switches, along with several other organizations' offices. Analyze both options in terms of network type, and evaluate which option is more suitable for global/large-scale communication, justifying your answer with at least two reasons. (5 Marks)
2	A school computer laboratory contains several computers connected through a network switch for sharing files, accessing learning resources, and communicating with a central server. Due to frequent movement of equipment, some network cables have become damaged, resulting in occasional transmission errors. When a computer sends data through the network, the switch receives the transmitted frames and checks them for errors. If a frame is found to be corrupted, the switch prevents the incorrect frame from being forwarded to the destination computer. A networking student is asked to examine the role of the Data Link Layer in maintaining reliable communication within the local network. Analyze the functions of the Data Link Layer in the above scenario and relate its role to error detection, framing, and delivery of data between connected devices. (5 Marks)
3	A packet is being transmitted from a source host to a destination host across multiple networks. Along the way, it must be: (a) routed based on the best available path, (b) uniquely addressed at the device/local-network level, and (c) transmitted as raw signals over the physical medium. Analyze this scenario using the OSI model to identify the three layers involved, explain the specific responsibility of each, and name the device typically associated with layer (a). (5 Marks)
4	(i) A small startup is setting up a local area network (LAN) to connect six employee laptops for sharing files, accessing printers, and using common network resources. Each laptop is connected to a central network switch using a separate dedicated cable. The switch manages communication between the connected laptops. If the cable connecting one laptop to the switch becomes damaged or disconnected, only that particular laptop loses network connectivity, while the remaining five laptops continue to communicate normally. The startup prefers this arrangement because it makes troubleshooting and network maintenance easier. Apply your knowledge of physical network topologies to identify the topology used in this scenario, and outline three characteristics of the identified topology that match the given situation. (3 Marks)
	(ii) During a live cricket match being streamed on a mobile application, the viewer notices that the video occasionally freezes or pauses for a fraction of a second. The audio may continue briefly, and then the video resumes after a short delay. The internet connection appears stable, and there is sufficient bandwidth available for streaming. This temporary interruption affects the smooth and continuous delivery of the live video. Identify the data communication characteristic responsible for this temporary freezing and define the characteristic. (2 Marks)
UNIT 1	**SECTION 2**

5	A network engineer is designing a burst-transmission protocol for a full-duplex link with a bandwidth of 5 bps and a delay of 5 seconds. The design requires the sender to send a burst of data and wait for acknowledgment before sending the next burst. Analyze the bandwidth-delay product for this link, and design an appropriate burst size (in bits) that makes full use of the full-duplex channel's capacity, justifying your calculation. (5 Marks)
6	A wireless Internet service provider (ISP) is planning a city-wide communication network with two different requirements. The first component must deliver a one-to-many broadcast service to thousands of homes distributed across the city, requiring wide coverage and simultaneous communication with multiple receivers. The second component must provide a dedicated point-to-point backhaul connection between two rooftop towers located 20 km apart, requiring focused transmission, high reliability, and minimum interference. The network designer is considering radio waves and microwaves for these two applications. Analyze the suitability of radio waves and microwaves for the two network components and justify the appropriate choice for each based on directionality, coverage, propagation characteristics, and interference. (5 Marks)
7	A city government is evaluating two wireless technologies to deploy free public information kiosks in libraries, municipal offices, and community centers. Technology X uses radio waves that can penetrate building walls, while Technology Y uses microwaves that require aligned directional antennas. Citizens report that Technology X occasionally causes cross-talk between kiosks located in adjacent rooms, resulting in one kiosk responding to signals intended for another. Analyze the root cause of the cross-talk problem and recommend whether switching to infrared communication for indoor kiosks would resolve the issue. Justify the recommendation based on penetration, directionality, coverage range, and interference characteristics. (5 Marks)
8	(i) A company conducts an important live video conference between employees working from different locations. During the call, voice and video packets are continuously transmitted across the network. The network administrator observes that the packets do not experience the same transmission delay. For example, one packet reaches the receiver after 20 ms, the next after 45 ms, and another after 40 ms. Although the packets are successfully delivered, their arrival times vary significantly. As a result, the audio and video may become uneven, delayed, or temporarily out of synchronization. Apply the concept of jitter to explain why this scenario is problematic for the video call application. (3 Marks)
	(ii) A wire carrying an electric signal becomes noticeably warm after being used for a while, and the received signal at the far end is weaker than what was originally sent. Identify the type of transmission impairment illustrated and explain its cause. (2 Marks)
UNIT 2	**SECTION 1**
9	A communication system transmits binary data over a noisy channel. The receiver must detect transmission errors before processing the received information. The engineer decides to use block coding. Apply the concept of block coding to show how it helps detect transmission errors and improves communication reliability using a suitable example. (5 Marks)
10	A banking application transfers customer transaction data as two 8-bit segments through a communication network. Before sending the packet, the sender generates a checksum to detect transmission errors. The receiver verifies the checksum before accepting the transaction.

	Apply the checksum algorithm to determine the checksum for the two data segments 10101010 and 11001100, and identify the checksum value that should be transmitted with the packet. (5 Marks)
11	The receiver obtains the following received codeword 1101011011000 using the generator polynomial 10011. After modulo-2 division, Calculate the CRC remainder using modulo-2 division and find whether the received frame contains transmission errors. (5 Marks)
12	(i) A receiver receives a data frame protected by CRC. Before accepting the frame, the receiver divides the received bits using the same generator polynomial used by the sender. The communication is considered successful only if the remainder is zero. Assess whether the received frame 11010110111110 is error-free using the generator polynomial 10011. (3 Marks)
	(ii) Apply the CRC encoding procedure for the data bits 1101011011 using the generator polynomial 10011, and determine the CRC bits. (2 Marks)
UNIT 2	**SECTION 2**
13	A communication engineer observes that the transmitted data contains the same byte pattern used as the frame delimiter in PPP. To prevent the receiver from misinterpreting the data, the sender modifies the transmitted frame before sending it. Apply the concept of byte stuffing to explain how PPP avoids incorrect frame interpretation during data transmission. (5 Marks)
14	A communication engineer observes that a PPP frame contains a data byte identical to the frame delimiter (0x7E). If transmitted without modification, the receiver may incorrectly interpret it as the end of the frame, resulting in communication failure. The engineer decides to apply an appropriate framing technique before transmission. Identify the technique used by PPP to handle special bytes in the Information field and explain how it ensures correct frame transmission. (5 Marks)
15	A networking student compares the Logical Link Control (LLC) and Medium Access Control (MAC) sublayers while studying IEEE 802 standards. The student wants to understand their responsibilities before implementing a LAN protocol. Compare the responsibilities of LLC and MAC sublayers and identify which sublayer is responsible for physical addressing. (5 Marks)
16	(i) A smart manufacturing plant connects hundreds of IoT sensors through both wired and wireless communication networks. The network architect must select appropriate medium access techniques to maximize throughput while minimizing transmission delays. Based on the scenario, Recommend suitable protocols for the wired and wireless segments of the network with appropriate justification. (3 Marks)
	(ii) Assess the role of multiple access protocols in supporting reliable communication over shared media. (2 Marks)
UNIT 3	**SECTION 1**
17	A regional Internet Service Provider (ISP) experiences severe congestion during evening hours because video streaming consumes most of the available bandwidth. Customers report poor streaming quality and

	frequent packet loss. Justify the measures required to reduce congestion and improve packet delivery. (5 Marks)
18	A company creates two VLANs using the following address ranges: • VLAN 10: 192.168.0.0/24 • VLAN 20: 192.168.0.128/24 Users experience intermittent connectivity, and duplicate IP addresses are detected. Contrast the existing IPv4 configuration with the corrected subnet configuration required to eliminate the addressing conflict. (5 Marks)
19	A financial institution experiences intermittent communication failures between two data centers. The network administrator decides to use multiple network diagnostic tools to identify the source of the problem. Compare the available network diagnostic tools to determine the appropriate tool for identifying the communication issue. (5 Marks)
20	(i) A logistics company transfers large files through a WAN connection whose Maximum Transmission Unit (MTU) is smaller than the transmitted packet size. The router must transmit the data successfully. (a) Construct the sequence of operations performed during IPv4 fragmentation. (3 Marks)
	(ii) (b) Outline the purpose of the Fragment Offset field. (2 Marks)